

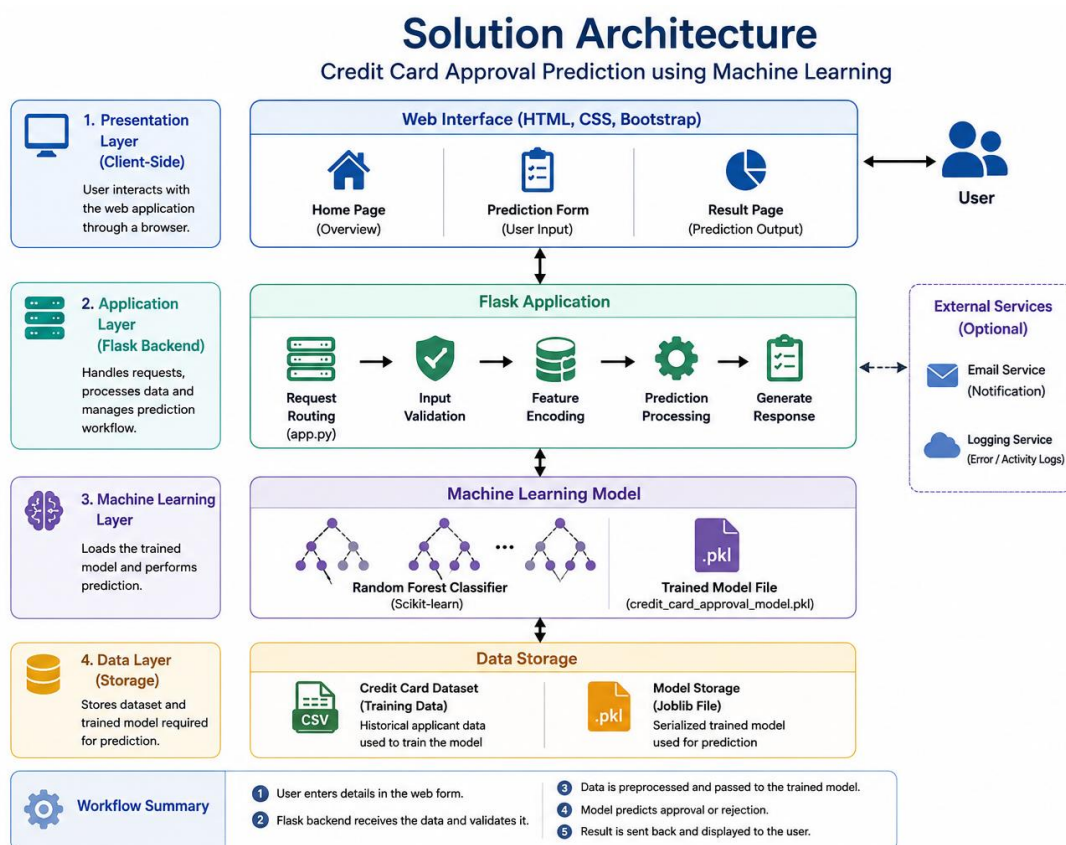
Solution Architecture

Date	02 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

The Solution Architecture provides a high-level view of the overall system structure and illustrates how different components interact to perform credit card approval prediction. It represents the flow of information from the user interface to the machine learning model and finally to the prediction result. The architecture ensures that each layer performs a specific responsibility, making the system modular, maintainable, and easy to enhance.

The proposed solution follows a layered architecture consisting of the **Presentation Layer**, **Application Layer**, and **Data Layer**. The Presentation Layer collects applicant information, the Application Layer validates and processes the data using the trained Random Forest model, and the Data Layer stores the trained model and dataset required for prediction.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for entering applicant information and displaying the prediction result.	HTML5, CSS3
Application Layer	Handles user requests, validates inputs, performs feature encoding, communicates with the machine learning model, and generates prediction results.	Python, Flask
Machine Learning Layer	Processes applicant information using the trained Random Forest model and predicts whether the credit card application will be approved or rejected.	Scikit-learn, Joblib
Data Layer	Stores the training dataset and the serialized machine learning model required for prediction.	CSV Dataset, Joblib (.pkl)